

Sterilization

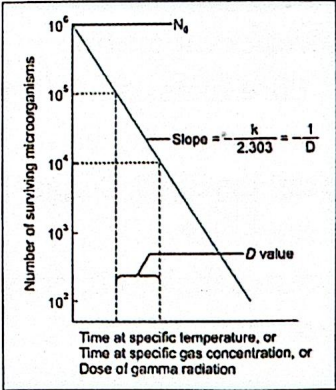
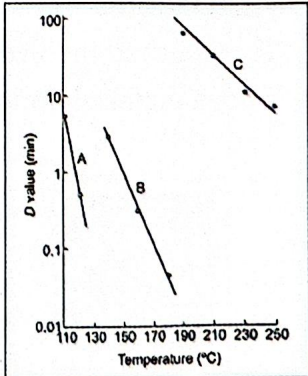
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INTRODUCTION

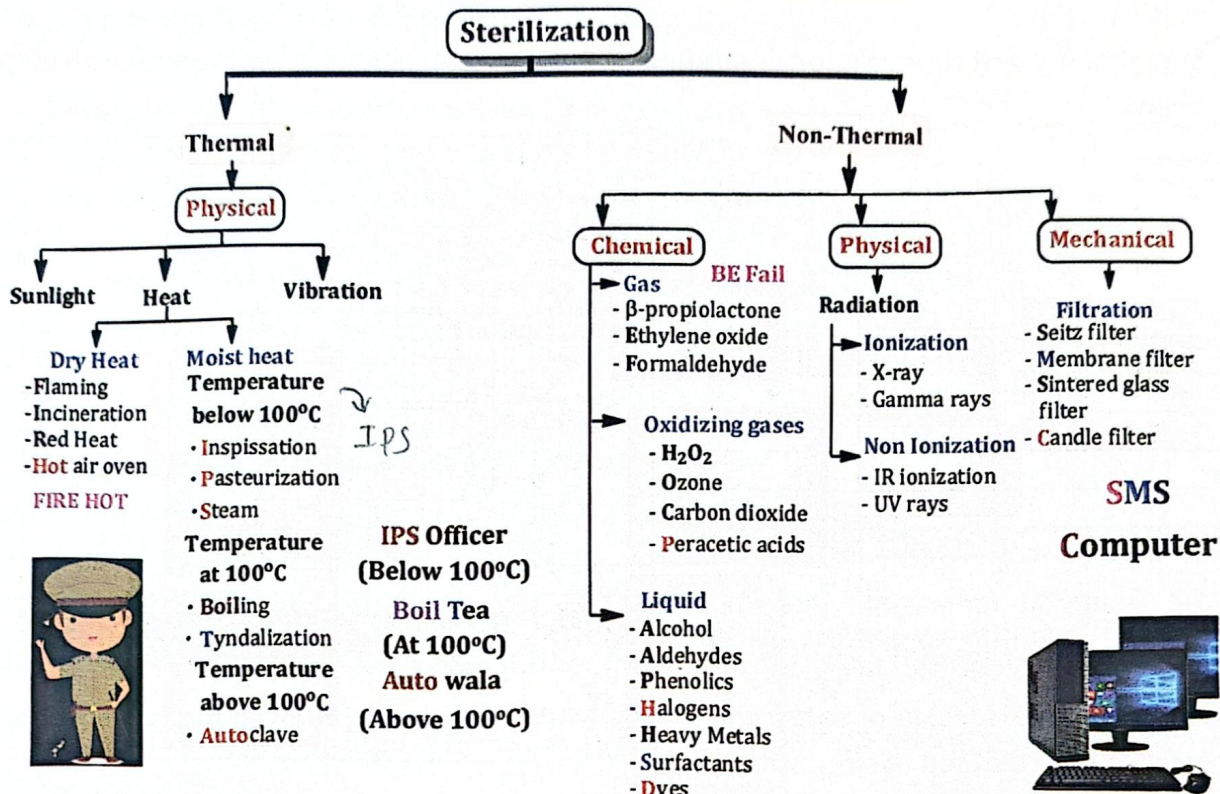
- **Sterilization** means **destruction of all living organisms** and their spores or their **complete removal from the preparation**.
- **Disinfection** is defined as a **process of destruction or removal of pathogenic microorganism/ kills vegetative cells** but not heat resistance spore from the object
- **Antiseptic** is a Chemical agent usually applied to living tissue in humans or animals in order to destroy harmful microorganism.
e.g. - Dettol, Phenol, Savlon

MICROBIAL DEATH KINETICS AND TERMINOLOGY [GPAT-2012, 2013, 2023]

D- Value (Decimal reduction time)	• The D-value is the time (for heat or chemical exposure) or the dose (for radiation exposure) required for the microbial population to decline by one decimal point (a 90%, or one logarithmic unit, reduction) $D = \frac{U}{\log N_0 - \log N_u}$ where, ✓ N_0 is the initial microbial population (product bio burden) ✓ N_u is the microbial population after receiving U time or dose units of sterilant exposure. 
Z- Value (Thermal resistance constant)	• Z-value describes the influence of temperature on decimal reduction time, D for microbial population. • It is the increase in temperature necessary to cause a 90% reduction in D value (1 log reduction). Z-value plots of log D versus temperature 

F- value (Thermal death time)	Time required to kill specific number of organism under defined condition.
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CLASSIFICATION OF STERILIZATION METHOD



MECHANISM OF ACTION OF STERILIZATION METHODS

STERILIZATION	PRINCIPLE
Dry heat (Hot air oven)	Thermal inactivation/Oxidation of cell constituents
Moist heat (Autoclaving)	Coagulation/Denaturation of Protein [GPAT-2012]
Radiation	Destruction/Disorganization of enzyme & DNA
Gaseous	Alkylation
Filtration	Physical separation of microorganism

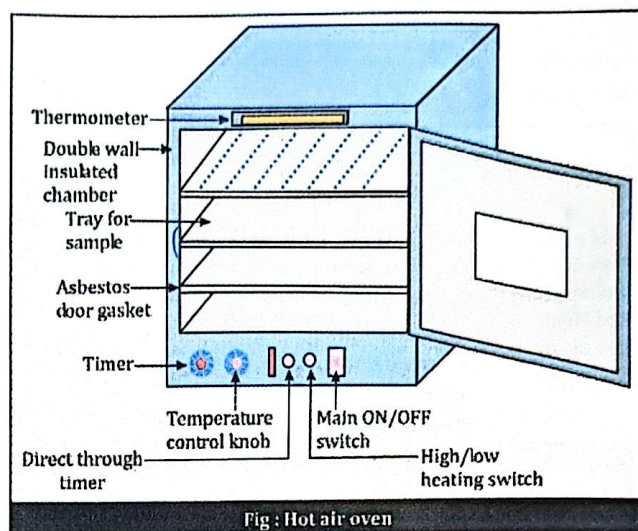
1. Thermal Physical Heat Sterilization

❖ Dry heat sterilization

- Used for **thermostable substances (above 140°C)**.
- The killing effect is due to protein denaturation, oxidative damage and the toxic effect of elevated levels of electrolyte.
- **Flaming**- The method is not reliable for sterilizing greasy or oily materials.
- **Incineration** - Instrument heated directly over flame.
- **Red heat**- Instrument heated directly over flame.

- **Hot air oven-** It works on the principle of the dry air sterilization process through **convection, conduction, and radiation**
- Substances that resist degradation at temperature between **140 to 260°C** may be sterilized by dry heat in hot air oven.
- **Method used for** - Petroleum Jelly, mineral oil, grease, waxes, talcum powder, glassware, Metalware etc.
- Temperature and time relationship required for sterilization in **hot air oven** is as given in table below

Temperature (°C)	Time (minutes)
170	60
160	120
150	150
140	180



❖ Moist heat sterilization

- Moist heat in the form of **saturated steam under pressure (autoclaving)** appears to be the most reliable method of sterilization.
- Killing effect is due to the **denaturation and coagulation of proteins**.

➤ Types

1. Temperature below 100°C
2. Temperature at 100°C
3. Temperature above 100°C

1. Temperature below 100°C *IPs*

✓ Pasteurization

- Pasteurization is developed by **Louis Pasteur**.
- It is a process in **packaged and non-packaged foods (such as milk and fruit juice)** are treated with **mild heat**, usually less than 100 °C (212 °F) to **eliminate pathogens** and extend shelf life.

TYPES	TEMPERATURES AND TIMES
Holder method	63°C for 30 min e.g. - Milk
Flash method or High temperature short time (HTST)	72°C for 15 second e.g. - Juice
Ultra-high temperature pasteurization	140°C for 3 second e.g. - Milk, Cream, Wine, Juice etc

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✓ **Inspissation –**

- Heating at **80-85°C** for **half an hour** daily on **three** consecutive days.
e.g.- Serum or egg media are sterilized.

2. **Temperature at 100°C**

- Tyndallization is also knowns **fractional sterilization/ intermittent sterilization**.
- Tyndallization- **100°C** for **10-30 min** for **3 successive days**.

3. **Temperature above 100°C**

- **Autoclaving** - The autoclave works on the principle of moist heat sterilization. Steam kills the microorganism by **coagulation or denaturation of protein**.

[GATE-2007]

- **Uses**- Autoclave is used sterilizing bacteriological media, heat stable liquid, saline, solution, heat resistance equipment, sealed ampoules containing thermostable drug solution, **glassware, syringe, rubber, surgical dressing and injection**.
- **Not used**- Autoclaving not used in anhydrous material such as **powder, oil, fat and Ointment**.

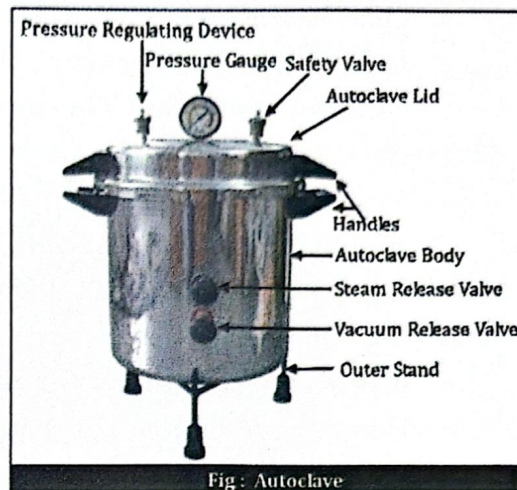


Fig : Autoclave

➤ Temperature/time/Pressure relationship for autoclave is as given in table below

MOIST HEAT STERILIZATION		
Autoclave		
Temp(°C)	Steam Pressure (psi)	Time (minutes)
115-116	10	30 min
121-123	15 [GATE-2007]	15 min [GATE-2007]
126-129	20	10 min
134-138	30	3 min

2. **Non Thermal Methods**

❖ **Chemical sterilization**

➤ **Gaseous sterilization**- **Destruction of all living microorganism** with chemical in gaseous or vapour state.

- Used for **Disposable plastic syringes** sterilization and made of polypropylene.
- Two newer gas used **Ethylene oxide & β -propiolactone**.

β -propiolactone

- β -Propiolactone is a cyclic lactone and is non-flammable liquid at room temperature.
- **Used in concentrations of 2-5 mg/L**.
- It is **25 times** more active disinfectant than formaldehyde gas.
- It is **4000 times** more active than ethylene oxide and **50000 times** more active than methyl bromide.
- β -Propiolactone is the most effective gaseous sterilizing agent.
- Due to **low penetration power and carcinogenicity**, not used for pharmaceutical applications.

Ethylene oxide

- Highly inflammable.
- It is mixed with **90% CO₂** and fluorinated hydrocarbons to make it non-inflammable. **[GATE-2008]**.
- Used at 55°C and pressure of 27 mm Hg and RH of 50-60%.
- Acts by alkylation of microbial metabolites.
- Chemically inert towards most **solid substance**.
- Used for medical & biological product such as **Catgut, Plastic, Books, Clothing, Antibiotic, cultured media and Thermolabile powders**.

Formaldehyde

- Formaldehyde gas is generated by adding **KMnO₄**. It reacts with vital biochemical compounds, such as **protein and nucleic acid**.
- Highly reactive chemically and difficult to remove from many materials after exposure.

Sulfur Dioxide

- It is highly toxic and carcinogenic; hence, **not used commonly**.
- It is known as formalin and contain (35 -40%) Formaldehyde. Formaldehyde is used for **fumigation of rooms**.

➤ Liquid sterilization

Alcohol

- Ethanol (60-70% v/v) & isopropanol (50-60%) used as skin disinfectant and methanol used as fungicidal

Halogens

- Chlorine and bromine **are strongly used**.
- **Germicidal action of chlorine is depend upon the formation of hypochlorous acid when free chlorine react with water.**

Heavy metal

- Mercury, silver and copper **are widely used as heavy metal**.

Phenol & its derivatives

- **Phenol act by mainly enzyme inactivation, disruption of cell, leakage of protein and precipitation of protein of microorganism.**
- Lysol **typically contains cresol, which is more germicidal than phenol**.

Quaternary ammonium compound

- **Most of cationic detergent are quaternary ammonium compound and it is act by disruption of cell wall & membrane.**

❖ Radiation Sterilization

- Radiation produces some heat when irradiated, hence, this method is also known as cold sterilization, **applicable to heat-sensitive materials**.
- It is based on wavelength & radiation power. Applied for hospital supplies, **vitamin, antibiotics, plastic syringes**.

- Act by damaging nucleic acid (DNA and RNA) of cells.
- These rays induce the production of abnormal nucleotides, such as thymine dimers.
- Based on the energy content, they are divided into 2 types-
 1. Non ionizing
 2. Ionizing

NON-IONIZING/UV RADIATION	IONIZING RADIATION
i) Infrared: For pre-packed items such as syringes and catheters. ii) UV radiation • UV rays in the region 253.7 Å for greatest activity in destroy microorganism • Its range is 150-3900 Å. • Used for aseptic filling rooms in industry, operation theaters and laboratories.	• Ionizing irradiation used for sterilization- i) γ (photon) irradiation: Most commonly used ii) β (electrons) irradiation • X-ray- x-ray have penetration ability to produce lethal effect • Gamma ray- gamma ray obtained by using radioactive isotope 10^4 Å to 10^{-4} Å e.g. - ^{60}Co (Cobalt) • Causes mutation & produces peroxide • Used for packed food material, Plastic material, Culture plate and Catheter, Antibiotics, Benzyl penicillin, disposable syringes, surgical catgut, bone and tissue grafts

❖ Mechanical Sterilization

➤ Filtration sterilization

- Depth filter are preferred for sterilization of **gases**.
 - Bacterial filter pore size **-0.45 μm- 0.2 μm**.
 - Testing of filters –**Bubble point test**.
 - It is determined the **surface tension** of the **liquid in capillary tubes**.
 - **High Efficiency Particulate Air (HEPA) filter-** used in operating rooms and burn units to remove bacteria from air.
 - Filtration is used for large volume of solution, **serum**, **eye drop** and **bacteriophage** and **toxin** etc.
- ✓ **Types of filters for sterilization [GPAT-2013, 2021]**

PARAMETERS	CONDITIONS
HEPA (High Efficiency Particulate Air Filters)[GPAT-2021]	
Efficiency	99.97 %
Particle size	0.3 μm[GPAT-2013]
Efficiency testing	DOP (Diocetyl phthalate) test
Conc. of DOP	66.6 ppm
Alternative to DOP	Liquid paraffin
Reagent used to check efficiency	Hydrocarbon vapour
Flow rate	20 ± 5 ft./min.

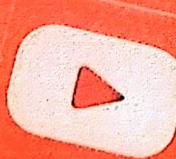
UEPA (Ultra Efficiency Particulate Air Filters) & VEPA (Very Efficiency Particulate Air Filters)	
Efficiency	99.997 %
Particle size	0.1 μm
Testing of HEPA filters	
Integrity testing (Once a year)	Cold DOP test
Airflow resistance	Pressure drops across filter (Manometer)

✓ Various bacterial proof filter for control use are

FILTER	CHEMICAL COMPOSITION
HEPA filter	Asbestos or glass
Millipore filter	Cellulose ester
Seitz filter	Asbestos or wood cellulose
Ceramic filter	Porcelain or Kieselguhr
Sintered glass filter	Borosilicate glass
Sintered metal filter	Stainless steel
Membrane filter	Cellulose acetate or cellulose nitrate
Asbestos	Magnesium silicate
Filter candle	Porous porcelain or Kieselguhr

STERILIZATION INDICATORS

STERILIZATION METHOD	INDICATOR
Dry heat	Biological indicators :- <i>Bacillus subtilis var niger</i> Chemical indicator :- Browne's tube, Bowie-Dick tape [GATE-2001, GPAT-2013]
Moist heat	Biological indicators:- Spores of <i>Bacillus stearothermophilus</i> , <i>Bacillus subtilis</i> , <i>Bacillus coagulans</i> and <i>Clostridium sporogenes</i>
Gaseous sterilization	Biological indicators :- <i>Bacillus subtilis var niger</i> [GATE-1999] Chemical indicator :- Royce sachet
UV radiation	Biological indicators- Spores of <i>Bacillus pumilus</i> , <i>Pseudomonas diminuta</i> also known as <i>Brevundimonas diminuta</i>
Filtration	Biological indicators- <i>Pseudomonas diminuta</i> or <i>Serratia marsence</i>



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PART B: STERILIZATION AND DISINFECTION

1. Historical Background

Ignaz Semmelweis (1847): Demonstrated life-saving potential of handwashing with chlorinated lime — earned title 'Father of Infection Control'.

Joseph Lister (1865): Applied carbolic acid (phenol) to wounds — revolutionized surgery by drastically reducing postoperative mortality.

2. Key Definitions and Terminology

STERILIZATION: The process by which an article, surface, or medium is freed of **ALL** living microorganisms, including **viable bacterial spores**. An **ABSOLUTE** concept. Sterilization implies destruction of spores (the most resistant microbial form, excluding prions).

DISINFECTION: The destruction or removal of **pathogenic organisms** capable of causing infection. Does NOT guarantee elimination of bacterial spores. A **relative** concept — graded into levels (High, Intermediate, Low).

ANTISEPSIS: Chemical disinfection of **living tissue** (skin, mucous membranes). Antiseptics must be non-toxic to host tissues.

DECONTAMINATION: A broad term — rendering an object safe to handle. Involves cleaning followed by disinfection or sterilization.

CLEANING: Physical removal of foreign material (organic matter, soil, dust) using water and detergents. **Most critical step in reprocessing** — without it, organic matter shields microorganisms from sterilants.

⚠ **EXAM FACT — Key Distinction:** Sterilization kills ALL organisms including SPORES. Disinfection kills PATHOGENS but NOT necessarily spores. Antiseptics are for LIVING tissue.

3. Hierarchy of Microbial Resistance

Order from MOST resistant to LEAST resistant:

Microorganism	Examples	Resistance Mechanism	Required Process
Prions	CJD, Kuru, Scrapie	Misfolded proteins, no nucleic acid	Special Prion Protocol
Bacterial Spores	B. anthracis, C. difficile	Thick spore coat, dipicolinic acid	Sterilization (Sporicidal)
Mycobacteria	M. tuberculosis, Cryptosporidium	Waxy mycolic acid cell wall	High-Level Disinfection
Non-enveloped Viruses	Poliovirus, Norovirus	Tough protein capsid (hydrophilic)	Intermediate Disinfection
Fungi	Candida, Aspergillus	Chitinous cell wall	Intermediate/Low Disinfection
Vegetative Bacteria	Staph, E.coli, Pseudomonas	Metabolically active	Low-Level Disinfection
Enveloped Viruses ↓ LEAST	HIV, HBV, Influenza, COVID	Lipid envelope easily disrupted	Low-Level Disinfection

⚠ **EXAM FACT:** HIV and HBV (enveloped viruses) are among the **EASIEST** to kill outside the body despite their clinical severity. MTB is the benchmark for Intermediate-Level Disinfection. Prions survive even standard autoclaving (121°C).

4. Spaulding's Classification of Medical Devices

Spaulding's Classification (Earle Spaulding): Gold standard for determining reprocessing requirements in clinical settings. Based on **risk of infection associated with device use**.

Critical Items (Sterile Site Contact)

- Enter sterile tissue or vascular system

- Requirement: **STERILIZATION (SAL 10^{-6})**
- Examples: **Surgical instruments, cardiac catheters, implants, needles, biopsy forceps**

Semi-Critical Items (Mucous Membrane Contact)

- **Contact mucous membranes or non-intact skin; do NOT penetrate sterile tissue**
- Requirement: **High-Level Disinfection (HLD)** — kills all microorganisms except HIGH numbers of bacterial spores
- Examples: **Flexible endoscopes, laryngoscopes, vaginal specula, anesthesia breathing circuits**

Non-Critical Items (Intact Skin Contact)

- **Contact only intact skin**
- Requirement: **Low-Level Disinfection (LLD)**
- Examples: **Stethoscopes, BP cuffs, bedpans, ECG leads**

⚠ **EXAM FACT** — Spaulding Mnemonic: Critical → Sterilization | Semi-critical → HLD | Non-critical → LLD

5. Physical Methods of Sterilization

Physical methods are **generally preferred** — reliable, no toxic residues, easy to monitor. Heat is the most important physical agent.

5.1 Sunlight and Drying

- UV component of sunlight has bactericidal activity — reduces microbial loads in water
- Unreliable for clinical sterilization
- **Desiccation:** Halts bacterial growth but does NOT kill them (spores survive desiccation for prolonged periods)

5.2 Dry Heat Sterilization

Mechanism: Kills microorganisms by **oxidation, charring, and protein denaturation**. Does NOT facilitate hydrolysis of peptide bonds. Therefore requires **higher temperatures and longer times** than moist heat.

Methods of Dry Heat

3. **Red Heat:** Hold article in Bunsen flame until red-hot. Uses: inoculating loops, wire tips in microbiology lab
4. **Flaming:** Pass article through flame briefly (not red-hot). Uses: mouthing of test tubes, glass slides
5. **Incineration:** Burns organic material to ash. Method of choice for pathological waste, animal carcasses, Yellow bag waste (under BMW Rules 2016)
6. **Hot Air Oven:** Most widely used dry heat method

Hot Air Oven — Standard Cycles

Temperature	Time (Holding)	Note
160°C	2 hours (120 min)	MOST COMMON Standard Cycle
170°C	1 hour (60 min)	Alternative
180°C	30 minutes	Alternative

Items Sterilized in Hot Air Oven

- **Glassware:** Petri dishes, flasks, pipettes, test tubes (must be perfectly dry)
- **Metals:** Forceps, scissors, scalpels (non-corrosive, unlike steam)
- **Non-aqueous substances — HIGH YIELD:** Liquid Paraffin, Dusting Powders (starch, talc), oils, fats, grease

⚠ **EXAM FACT:** Moist heat **CANNOT** sterilize oils/powders — steam cannot penetrate them. Therefore **Liquid Paraffin, Talc, Starch, Oils = HOT AIR OVEN** is the only method.

Hot Air Oven Monitoring

- **Biological Indicator:** Spores of **Bacillus atrophaeus** (formerly *B. subtilis* var. niger)
- **Chemical Indicator:** **Browne's tubes** (Green spot) — change from red to green on adequate exposure
- **CRITICAL SAFETY:** Do NOT open oven door immediately — allow temperature to fall below 60°C to prevent thermal shock cracking of glassware

5.3 Moist Heat Sterilization

Mechanism: Kills microorganisms by **denaturation and coagulation of proteins**. Superior to dry heat because: (1) **Latent heat of vaporization** (steam releases 540 cal/g on condensation); (2) **Better penetration** (steam volume contracts 1/1600th on condensation, creating negative pressure drawing steam deeper); (3) **Hydrolysis** of peptide bonds at lower temperatures.

Methods Below 100°C

Pasteurization:

- **Holder Method:** 63°C for 30 minutes
- **Flash Method (HTST):** 72°C for 15–20 seconds
- Kills: *M. tuberculosis*, *Salmonella*, *Coxiella burnetii* (most heat-resistant non-sporing pathogen in milk). Does NOT kill spores.

⚠ **EXAM FACT:** *Coxiella burnetii* is the benchmark organism for pasteurization — it is the most heat-resistant non-sporing pathogen in milk. Pasteurization does NOT achieve sterilization (spores survive).

Inspissation:

- For culture media with high protein content (serum, egg) that would coagulate at higher temperatures
- Examples: **Lowenstein-Jensen (LJ) Medium** (for TB), **Loeffler's Serum Slope**
- Protocol: 80°C–85°C for 30 minutes on 3 consecutive days

Methods at 100°C

Boiling:

- 100°C — kills vegetative bacteria within **10–30 minutes**
- **NOT** a reliable sterilization method — some bacterial spores survive boiling for hours
- **Method of disinfection, NOT sterilization**
- Adding **2% Sodium Bicarbonate (NaHCO₃)** promotes sterilization and prevents corrosion of metal instruments

Tyndallization (Intermittent Sterilization / Fractional Sterilization):

- Used for media containing **sugars or gelatin** that might decompose at autoclave temperatures
- Protocol: **Steam at 100°C for 20 minutes on 3 successive days**

Day 1: Kills vegetative cells; spores survive but are heat-shocked

- Overnight incubation: Spores germinate into vegetative cells
- Day 2: Kills newly germinated vegetative cells
- Day 3: Kills remaining spores

Methods Above 100°C: The Autoclave

Principle: By trapping steam in a pressurized chamber, temperature rises above 100°C. At **15 psi above atmospheric pressure**, pure saturated steam reaches **121°C**.

Standard Cycle Parameters:

- Temperature: **121°C (250°F)**
- Pressure: **15 psi (1.05 kg/cm²)**
- Holding Time: **15–20 minutes**

⚠ **EXAM FACT — AUTOCLAVE:** **121°C / 15 psi / 15–20 minutes**. Flash autoclave: **134°C / 30 psi / 3–5 minutes**. Flash is also used for Prion decontamination (extended to 18 minutes at 134°C).

Types of Autoclaves:

7. **Gravity Displacement:** Steam enters top; forces air (heavier) out through drain at bottom. Slower; used for simple loads (media, glassware)
8. **Pre-vacuum (Pulsed Vacuum):** Vacuum pump removes air BEFORE steam admission. Rapid and deep penetration of porous loads (surgical packs, linens). Tested by Bowie-Dick test

Articles Sterilized by Autoclave:

- Surgical instruments (except delicate/sharp microsurgical ones)
- Dressings, linens, gowns, culture media (Nutrient Agar, MacConkey Agar)
- **Biomedical Waste (Red Bag):** Syringes, gloves, catheters, plastics — autoclaved before recycling
- **Biomedical Waste (Yellow Bag — Labs):** Microbiology cultures pre-treated by autoclaving

Autoclave Failures:

- **Air Entrapment:** Most common cause of failure — air pockets prevent steam contact, lower temperature
- **Wet Packs:** Insufficient drying → packs remain wet → wick bacteria → considered **NON-STERILE**

Autoclave Monitoring:

- **Biological Indicator (Gold Standard):** Spores of Geobacillus stearothermophilus (thermophilic, grows at 55°C) — failure to grow after cycle confirms sterilization
- **Chemical Indicators:**
 - External (Class 1): **Autoclave Tape** — white stripes turn black; shows pack processed (NOT sterility)
 - Internal: Chemical strips placed inside packs to verify steam penetration
 - Bowie-Dick Test (Class 2): **Dense pack test** — tests for air leaks and steam penetration in PRE-VACUUM autoclaves only

5.4 Filtration

Principle: Removes microorganisms physically WITHOUT killing them. Only method for sterilizing heat-labile liquids (sera, antibiotic solutions, vaccines, sugar solutions) and gases.

Types of Filtration

- **Membrane Filters (Cellulose esters):** Screening/sieving mechanism. Standard pore size: 0.22 µm or 0.45 µm for bacteria
- **Depth Filters (glass fibers):** Adsorption/entanglement mechanism. Particles trapped in tortuous channels

HEPA Filters (High-Efficiency Particulate Air)

- Used to sterilize air in Operation Theaters, Laminar Flow Hoods, and Biological Safety Cabinets
- **Specification:** Removes **99.97%** of particles **0.3 µm** in diameter
- **0.3 µm = Most Penetrating Particle Size (MPPS)** — particles both larger and smaller are actually easier to trap
- Mechanism: Interception, impaction, and diffusion

△ **EXAM FACT:** HEPA filter removes 99.97% of 0.3µm particles. 0.3µm is the HARDEST particle to trap (MPPS). Used in OT, laminar flow, BSC.

5.5 Radiation Sterilization

Ionizing Radiation ('Cold Sterilization')

Mechanism: Ionization of water molecules → highly reactive free radicals (hydroxyl, superoxide) → single and double-strand **DNA breaks**.

- **Sources:** Gamma Rays (from Cobalt-60 or Cesium-137) — high penetration power; Electron Beams (E-beam) — lower penetration, higher dose rate
- Applications: Industrial sterilization of disposable medical items — plastic syringes, catheters, IV sets, grease, suture materials (catgut), bone grafts
- Called 'Cold': **generates negligible heat** — ideal for thermolabile plastics
- **Biological Indicator:** Bacillus pumilus

Non-Ionizing Radiation (Ultraviolet)

- **Source:** Mercury vapor lamps
- **Wavelength:** Germicidal range 240–280 nm; **254 nm = optimal**
- **Mechanism:** UV absorbed by DNA → forms Thymine-Thymine (TT) dimers → distorts DNA helix → inhibits replication
- **Limitation:** **Poor penetration** — cannot penetrate glass, paper, or dirt. Surface and air disinfection **ONLY**
- Applications: Air disinfection in OTs, surfaces of Laminar Flow Hoods/Biosafety Cabinets

△ **EXAM FACT:** UV radiation — Germicidal wavelength = **254 nm**. Causes **Thymine-Thymine dimers**. UV **CANNOT** penetrate glass, paper, or organic matter. Only for surface/air use. **No penetration into wrapped items.**

6. Chemical Methods of Disinfection and Sterilization

6.1 Alcohols

- Agents: Ethyl alcohol (Ethanol) and Isopropyl alcohol (IPA)
- Mechanism: Denaturation of proteins + dissolution of lipid membranes

The Concentration Paradox — HIGH YIELD

- **70% Alcohol in water = MOST EFFECTIVE** — penetrates cell wall, coagulates proteins
- **100% (Absolute) Alcohol = LESS EFFECTIVE** — dehydrates cell surface rapidly, 'fixes' proteins as protective barrier preventing further penetration

- **Spectrum:** Bactericidal, Tuberculocidal, Fungicidal, Virucidal (enveloped viruses like HIV, HBV). **NOT sporicidal**
- **Uses:** Skin antiseptics, clinical thermometers, rubber stoppers of vials, stethoscope diaphragms

⚠ **EXAM FACT:** 70% alcohol > 100% alcohol for disinfection. Water is required to denature proteins. 100% absolute alcohol actually **PRESERVES** (fixes) bacteria.

6.2 Aldehydes

Mechanism: ALKYLATION — replacing hydrogen atoms on amino, carboxyl, sulfhydryl, hydroxyl groups of proteins and nucleic acids with alkyl groups → cross-linking and disruption.

Glutaraldehyde (Cidex) — High-Level Disinfectant

- **Formulation:** 2% acidic solution (stable) — must be '**activated**' by **alkalinization** (pH 7.5–8.5)
- Once activated: shelf life **14–28 days**
- **Protocol for Endoscopes:**
 - **HLD:** Immersion for **20 minutes** at 20°C
 - **Sterilization:** Immersion for **10–12 hours**
- **Uses:** Standard agent for **heat-sensitive endoscopes** (Bronchoscopes, Gastrosopes), Cystoscopes, anesthetic equipment
- **Safety:** Respiratory irritant and sensitizer — proper ventilation required

Ortho-phthalaldehyde (OPA)

- Newer aldehyde (**0.55%**) — rapidly replacing glutaraldehyde
- **Advantages:** No activation required; faster HLD (**12 minutes**); better mycobactericidal; low vapor (less irritation)
- **Disadvantage:** Stains proteins (skin, mucous membranes, clothing) **GRAY**

Formaldehyde

- **Liquid (Formalin):** 37–40% solution. Used for **preserving anatomical specimens**, killing bacterial cultures for vaccine preparation (toxoids)
- **Gas:** Used for fumigation of OTs (historically) and sterilization of filters. Requires high humidity
- **Toxicity:** Highly **carcinogenic** and irritant

6.3 Halogens

Chlorine (Hypochlorites)

Mechanism: Potent oxidizing agent. In water, forms **Hypochlorous acid (HOCl)** which enters cells and **oxidizes enzymes**. HOCl is much more active than hypochlorite ion (OCl^-).

- **Uses:** Disinfection of water, surfaces, and large organic spills

Spill Management Protocol — HIGH YIELD

Situation	Sodium Concentration	Hypochlorite	Contact Time
General Surface Disinfection	1% (1,000 ppm)		10–20 min
Large Blood/Body Fluid Spills	10% (10,000 ppm)		20 minutes
Prion Decontamination	20,000 ppm Chlorine OR 1N NaOH		1 hour

⚠ **EXAM FACT:** Blood spill → **10% Sodium Hypochlorite (10,000 ppm)** for 20 minutes. **DO NOT** use chlorine on urine spills — releases toxic chlorine/chloramine gas. Cover spill with absorbent material **FIRST**.

Iodine

- **Mechanism:** Oxidizing agent; iodates proteins
- **Tincture of Iodine:** 2% Iodine in 70% Alcohol
- **Iodophores (Povidone-Iodine/Betadine):** Iodine complexed with PVP. Releases iodine slowly, reducing irritation and staining. Used for surgical hand scrubbing and skin prep

6.4 Oxidizing Agents

Hydrogen Peroxide (H_2O_2)

- **Mechanism:** Production of **hydroxyl free radicals ($\text{OH}^\bullet$)** → attack membrane lipids and DNA

- Uses: 3% for wound cleaning; 6–25% for chemical sterilization; **Gas Plasma sterilization (STERRAD)**

Peracetic Acid (PAA)

- Effective **sporicidal agent** even at low temperatures
- **Mechanism:** Denaturation and oxidation
- Uses: **Automated Endoscope Reprocessors (STERIS system)**
- **Advantage:** Breaks down into acetic acid (vinegar) and oxygen — **environmentally safe**

7. Gaseous and Low-Temperature Sterilization

7.1 Ethylene Oxide (ETO)

Mechanism: ALKYLATION — replaces hydrogen atoms with hydroxyethyl groups in microbial DNA and proteins → blocks replication.

Parameter	Value
Gas Concentration	450–1200 mg/L
Temperature	37°C–63°C (Low temperature)
Humidity (CRITICAL)	40%–80% — essential to hydrate spore coat and allow gas penetration
Exposure Time	1–6 hours
Aeration (CRITICAL)	8–12 hours at 50–60°C to remove toxic residual ETO
Biological Indicator	<i>Bacillus atrophaeus</i>

- Uses: Heart-lung machine, sutures, dental equipment, respirators, complex devices with **long lumens**. Penetrates packaging (plastic/paper) well

⚠ **EXAM FACT: ETO** — Low-temperature sterilant. Humidity (40–80%) is **CRITICAL** for efficacy. Requires long aeration. **ETO is carcinogenic and flammable** — must be used in specially designed chambers.

7.2 Plasma Sterilization (STERRAD — Hydrogen Peroxide Gas Plasma)

- **Technology:** H₂O₂ vapor + radiofrequency energy → plasma state
- **Mechanism:** Generation of free radicals (hydroxyl, hydroperoxyl) → destroy cell components

Key Advantages

- **Rapid cycle:** 28–75 minutes
- **NO toxic residues:** Byproducts are non-toxic O₂ and H₂O vapor. No aeration required
- **Low temperature (< 50°C):** Ideal for delicate electronics

Applications and Limitations

- Preferred method for **modern endoscopes, cameras, light cords, power drills**
- **CANNOT** process cellulose (paper/linens), liquids, or powders
- **Biological Indicator:** *Geobacillus stearothermophilus*

8. Sterilization Method Summary Table

Item	Best Method	Biological Indicator	Notes
Glass Syringes	Hot Air Oven	<i>B. atrophaeus</i>	Alt: Autoclave
Disposable Syringes Plastic	Gamma Radiation	<i>B. pumilus</i>	Alt: ETO
Surgical Instruments	Autoclave	<i>G. stearothermophilus</i>	Alt: Plasma/Dry Heat
Culture Media (General)	Autoclave	<i>G. stearothermophilus</i>	—
Culture (Serum/Egg) Media	Inspissation	—	80–85°C × 3 days

Liquid Paraffin / Oils / Powders	Hot Air Oven ONLY	B. atrophaeus	No alternative
Suture Materials (Catgut)	Ionizing Radiation	B. pumilus	—
Flexible Endoscopes	2% Glutaraldehyde HLD	—	Alt: Plasma/OPA
Sera / Vaccines / Antibiotics	Filtration (0.22 µm)	B. diminuta	Heat-labile

9. Biomedical Waste Management (BMWM) Rules 2016

BMWM Rules 2016: Strictly regulate biomedical waste in India. Mandate **segregation at source** into four color-coded categories. **HIGH YIELD** for all public health competitive exams.

9.1 The Four Color Categories

Color	Type of Waste	Treatment	Key Examples
YELLOW	Anatomical, Infectious, Chemical waste	Incineration / Plasma Pyrolysis	Tissues, organs, soiled dressings, expired drugs, cytotoxic drugs, blood bags, microbiology cultures, live vaccines
RED	Contaminated RECYCLABLE Plastics	Autoclave/Microwave + Shredding	Syringes (no needle), IV tubing, gloves, catheters, <u>urine bags</u> , vacutainers
WHITE	Waste Sharps (Metals)	Autoclave or Dry Heat + Shredding	Needles, syringes with fixed needles, scalpels, blades, suture needles
BLUE	Glassware and Metallic Implants	Disinfection/Autoclave + Recycling	Broken glass, vials, ampoules, metallic implants, plates, screws

9.2 High-Yield Exam Scenarios

Blood Bags vs Urine Bags

⚠ CRITICAL: Blood Bags = YELLOW bin (anatomical/soiled waste — pre-treat by autoclaving/disinfection FIRST). Urine Bags = RED bin (plastic recyclable waste). Frequently tested distinction!

Vaccine Vials

- Live Attenuated Vaccines (Measles, OPV): Discard in **YELLOW bag** (treat as infectious — autoclave first if possible)
- Killed/Subunit Vaccines: Glass vial → **BLUE box**

Syringes

- Use **hub cutter** to separate needle from barrel
- Needle → **WHITE container** | Plastic barrel → **RED bin**

Cytotoxic Drugs

- Expired cytotoxic drugs returned to manufacturer OR incinerated at **>1200°C**
- Separate **YELLOW bag** labeled 'Cytotoxic'

10. Prion Decontamination Protocol

⚠ EXAM FACT: Prions resist standard autoclaving (121°C) and most chemical disinfectants. **Special protocol required:**

9. Immerse in 1N Sodium Hydroxide (NaOH) OR Sodium Hypochlorite (20,000 ppm) for 1 hour
10. Follow with Autoclaving at 134°C for 18 minutes (or 132°C for 1 hour)
11. If instrument cannot withstand protocol → **Incinerate**

11. Practical Protocol: Endoscope Reprocessing (7 Steps)

Flexible endoscopes — **Semi-critical items** — require HLD minimum. Complex, heat-sensitive, prone to biofilm.

12. **Pre-cleaning:** Wipe exterior and flush channels with detergent immediately after use
13. **Leak Testing:** Ensure scope is watertight
14. **Manual Cleaning:** Brush channels with enzymatic cleaner — **MOST IMPORTANT STEP**
15. **High-Level Disinfection:** 2% Glutaraldehyde for 20 min OR OPA for 12 min; AERs using Peracetic Acid also used
16. **Rinsing:** Rinse with sterile or filtered water to remove toxic residues
17. **Drying:** Flush channels with 70% alcohol + forced air — prevents **Pseudomonas/Acinetobacter** growth during storage
18. **Storage:** Hang vertically to facilitate drying

12. Monitoring and Quality Control of Sterilization

Sterilization Method	Biological Organism	Indicator	Rationale
Steam (Autoclave)	Geobacillus stearothermophilus	Thermophilic (55°C growth); highly resistant to moist heat	
Dry Heat (Hot Air Oven)	Bacillus atrophaeus	Resistant to desiccation and dry heat	
Ethylene Oxide (ETO)	Bacillus atrophaeus	Resistant to alkylating agents	
Plasma (STERRAD)	Geobacillus stearothermophilus	H ₂ O ₂ resistance	
Ionizing Radiation	Bacillus pumilus	High radiation resistance	
Filtration	Brevundimonas diminuta	Small size (0.3 μm); tests pore retention (0.2 μm filter)	

⚠ **EXAM MNEMONIC — Biological Indicators:** Autoclave/Plasma/~~Dry Heat (ETO)~~ = *Geobacillus stearothermophilus* (G--) | Hot Air Oven/ETO = *B. atrophaeus* | Radiation = *B. pumilus* | Filter = *B. diminuta*